Controlling Information Product Quality

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Introduction

- Tools used to control the quality of output in a manufacturing process can be used to control the quality of information products.

- In order to apply these TQM principles to DQ, methods must be adapted for measuring quality.

Quality Assurance

- Quality Assurance – Ensuring that quality requirements will be fulfilled.
- Focuses on processes and procedures necessary to produce a quality product or service.
- Proactive and preventive – steps taken BEFORE a product is produced

Quality Control

- Quality control – Monitoring specific outcomes to determine if they comply with relevant quality standards and identifying ways to eliminate causes of unsatisfactory performance.

- Measures quality AFTER the product is produced.

Acceptance Sampling

- Acceptance Sampling – Used to determine whether to accept or reject a lot based on pre- specified thresholds.
- Questions – What, where, and how much to inspect.
- Uses one of the four methods of ransom sampling – pure, systematic, stratified, clustered

What to Inspect

- Must be done in consultation with primary users, identify key quality characteristics for each information product (IP)

- Special concern: What are the quality defects that could significantly impair product performance
(degrade the value in its application) –
trace back to individual data requirements

Where to Inspect

- Map the flow of data through system.

- Identify junctures where data is most likely to change or be damaged, especially where it is
  - Crossing system boundaries
  - Crossing organization boundaries

How Much To Inspect

- Inspection can be costly!
- Use an economic model of the cost/benefits of inspection alternatives to determine where to best apply resources
- Stratified sampling can be the most effective when the defect rate is low

Trade-Offs of Acceptance Sampling

- Advantages:
  - Relatively inexpensive
  - Reduces inspector fatigue vs. 100% inspection
  - 100% may not be possible, may require destroying the product
- Disadvantages:
  - Some defective products may not be detected
  - Inspection does not prevent poor quality, it only detects it
  - Inspection is not a value-added activity (overhead)

Activities that Lead to Improved Quality

- Design of experiments
- Pareto diagrams
- Control charts
- Six sigma

Design of Experiments

- When creating an IP, experiments can be designed to probe the sensitivity of the output to fluctuations in the quality of the raw data inputs

- Similar to manufacturing experiments designed to determine the sensitivity of key characteristics of a product to changes in the parameters of the production process

Example: Transformer Based Matching

Record_x	Record_y	score
John R. Smith,123 Oak St, Little Rock, AR 72212	John R. Smith, 123 Oak St, Little Rock, AR 72212	1
John R. Smith,123 Oak St, Little Rock, AR 72212	John Smith,123 Oak St, Little Rock, AR 72212	0.984069
John R. Smith,123 Oak St, Little Rock, AR 72212	Smith, John R., 123 Oak St, Little Rock, AR 72212	0.977994
John R. Smith,123 Oak St, Little Rock, AR 72212	J. R. Smith,123 Oak St, Little Rock, AR 72212	0.930138
John R. Smith,123 Oak St, Little Rock, AR 72212	J R Smith,123 Oak St, Little Rock, AR 72212	0.925592
John R. Smith,123 Oak St, Little Rock, AR 72212	Jon Smith,123 Oak St, Little Rock, AR 72212	0.915689
John R. Smith,123 Oak St, Little Rock, AR 72212	Robert Woods, 456 Elm St, Little Rock, AR 72212	0.760728
John R. Smith,123 Oak St, Little Rock, AR 72212	Barry Sims, 78 Pine Ave, Little Rock, AR 72212	0.571927
Larry Sims, 78 Pine Ave, Little Rock, AR 72212	Barry Sims, 78 Pine Ave, Little Rock, AR 72212	0.856745
Larry Sims, 78 Pine Ave, Little Rock, AR 72212	J. R. Smith,123 Oak St, Little Rock, AR 72212	0.633224
Larry Sims, 78 Pine Ave, Little Rock, AR 72212	J R Smith,123 Oak St, Little Rock, AR 72212	0.632793
Larry Sims, 78 Pine Ave, Little Rock, AR 72212	Jon Smith,123 Oak St, Little Rock, AR 72212	0.624991

Pareto Charts

- Vertical bar graph that displays data defects according to the frequency of occurrence by category of data defect
  - From highest to lowest frequency

- Pareto's Rule
  80% of the problems stem from 20% of the causes

- Pareto analysis allows users to easily assign priorities for their efforts

Pareto Chart Structure

- Pareto chart has two Y-axes:
  - Left Y-axis tracks raw frequencies
  - Right Y-axis tracks cumulative percentages across categories
- Superimposes the cumulative percentage curve over the raw frequency bars by using the two separate Y-axes.
- Allows users to make inferences about the impact of correcting the worst problems

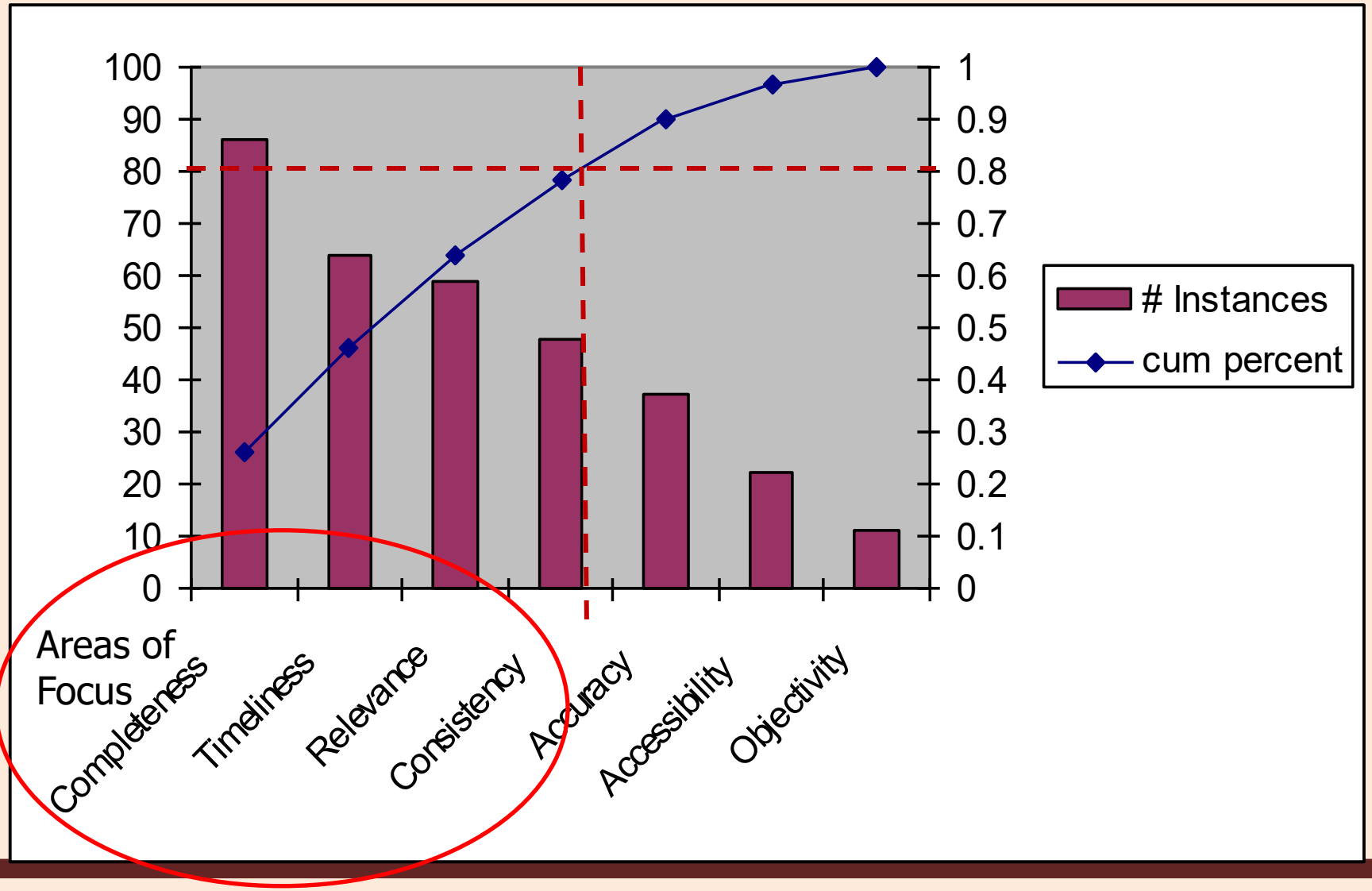
Example: Data Quality Problems

Dimension	# Instances
Accuracy	37
Completeness	86
Timeliness	64
Consistency	48
Objectivity	11
Relevance	59
Accessibility	22

Data Organized for Pareto Chart (Descending Order by Frequency)

Dimension	# Instances	Cumulative percent
Completeness	**86**	**0.26**
Timeliness	**64**	**0.46**
Relevance	**59**	**0.64**
Consistency	**48**	**0.79**
Accuracy	**37**	**0.90**
Accessibility	**22**	**0.97**
Objectivity	**11**	**1.00**

Pareto Chart



Control Charts

- Can be used to study the output of any repetitive process
- The primary tool of Statistical Process Control (SPC)
- Also called a Shewhart Chart after its inventor Walter A. Shewhart (Bell Labs, 1920s)

Purposes of Control Charts

1. Quantify the goal or standard the process should attain
2. Help the process achieve that goal
3. Check whether the goal has been attained

Control Chart Theory

There are two components of variation in quality

1. Variation due to **assignable cause** also called **special cause**

2. Variation due to **chance** also called random variation or **common cause**

In-Control Process

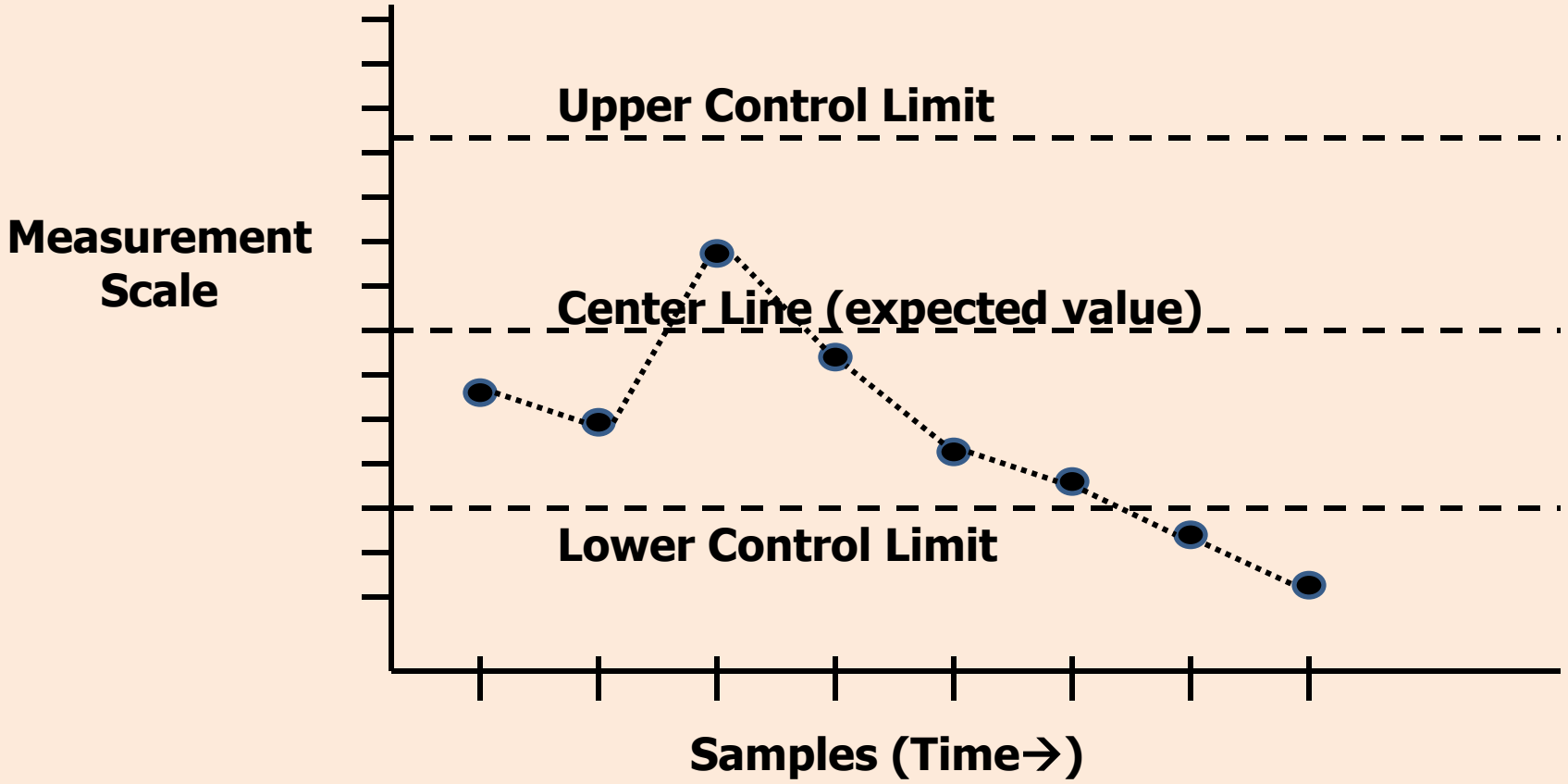
- Variances that cannot be traced to a particular cause are assumed to be due chance or simple random variations

- A process is said to be "**in control**" when variation in data is due to chance alone, i.e. common cause

Out-of-Control Process

- Variances that can be traced to a special cause
- Variation that exceeds a threshold of what is expected from chance
- A process is said to be "**out of control**" when variation exceeds the chance variation threshold and assumed to be due to special or assignable causes

General Control Chart Structure



Example Rules for Out-of-Control Processes

- There are 8 consecutive measurements above the mean or below the mean
- There are 8 consecutive measurements that are monotonically increasing or decreasing.
- 2 of 3 consecutive measurements above a 2σ limit or 2 of 3 below a -2σ limit
- 4 of 5 consecutive measurements are above a 1σ limit or 4 of 5 are below a -1σ limit

Describing Control Chart Decisions

Chart Says	If Common Cause	If Special Cause
In Control	**Correct decision**	**Failure to detect**
Out of control	**False alarm**	**Correct decision**

Two Categories of Control Charts

- Variables Charts
  - Monitor quantitative characteristics (actual measurements, continuous values)
  - Individual measurements, sample mean, sample range
- Attributes Charts
  - Monitor qualitative characteristics (discrete values)
  - Number of defects, proportion of defects

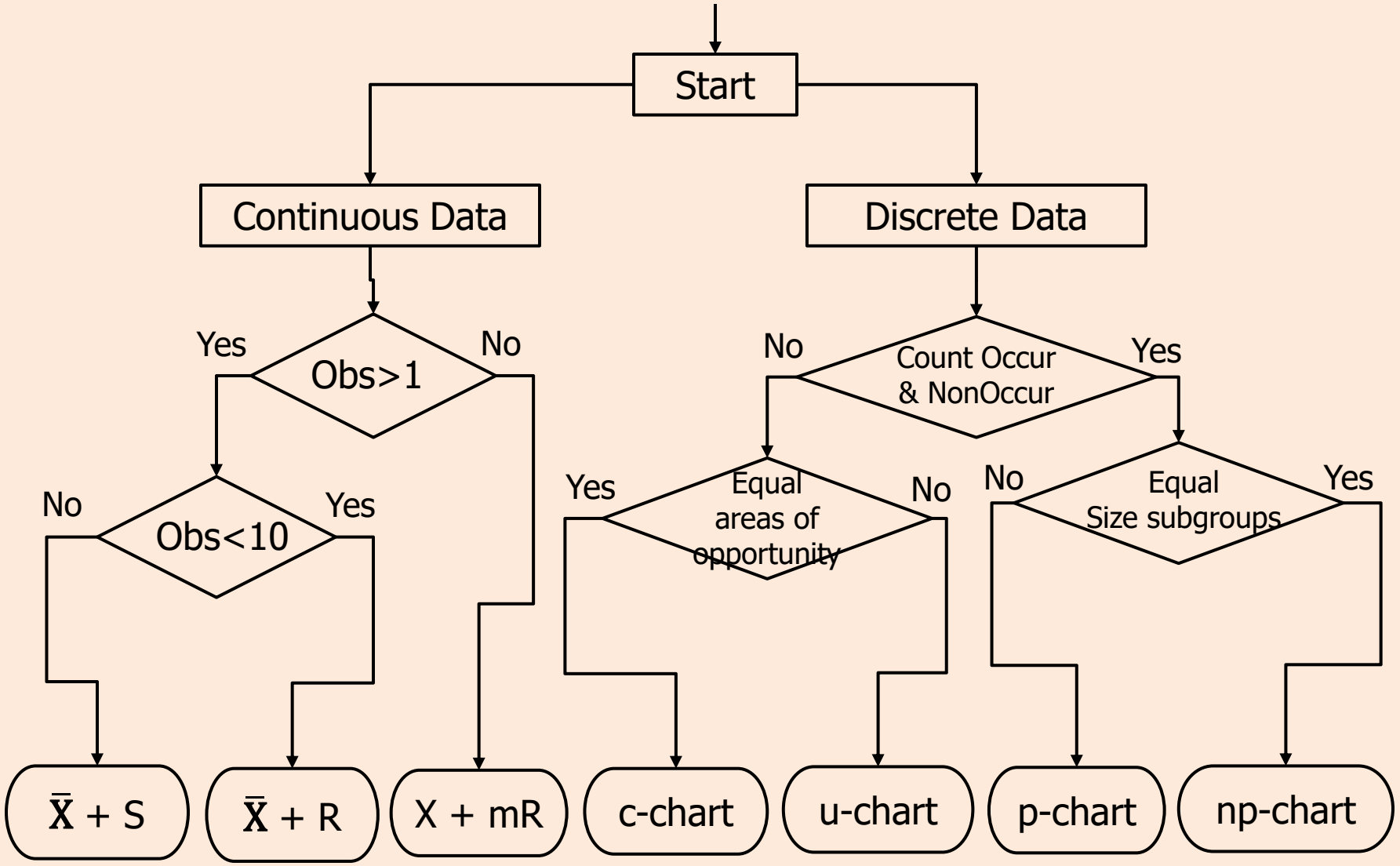
Types of Variables Charts

Chart Type	Data
X-bar	Sample means are plotted
R	Sample ranges are plotted
S^2 or s	Sample variances or standard deviations are plotted
X	Individual values are plotted
MR	Moving ranges are plotted

Type of Attributes Charts

Chart	Data	Comments
np	Number of defectives	Binomial distribution
p	Proportion of defectives	Can use varying sample sizes
c	Number of defectives per unit	Poisson distribution
u	Rate of defectives per unit	Can use varying sample sizes

Deciding Type of Chart



Centerline and Limits for a Number Non-conforming (np) Chart

Suppose there are "k" samples all of size "n"

Let X_i represent the number of non-conforming items in the i-th sample

$$\text{Center Line} = n\bar{p} = \frac{\sum_i X_i}{k}$$

$$\text{Proportion of Defects} = \bar{p} = \frac{n\bar{p}}{n}$$

$$\text{Control Limits} = n\bar{p} \pm 3\sqrt{n\bar{p}(1-\bar{p})}$$

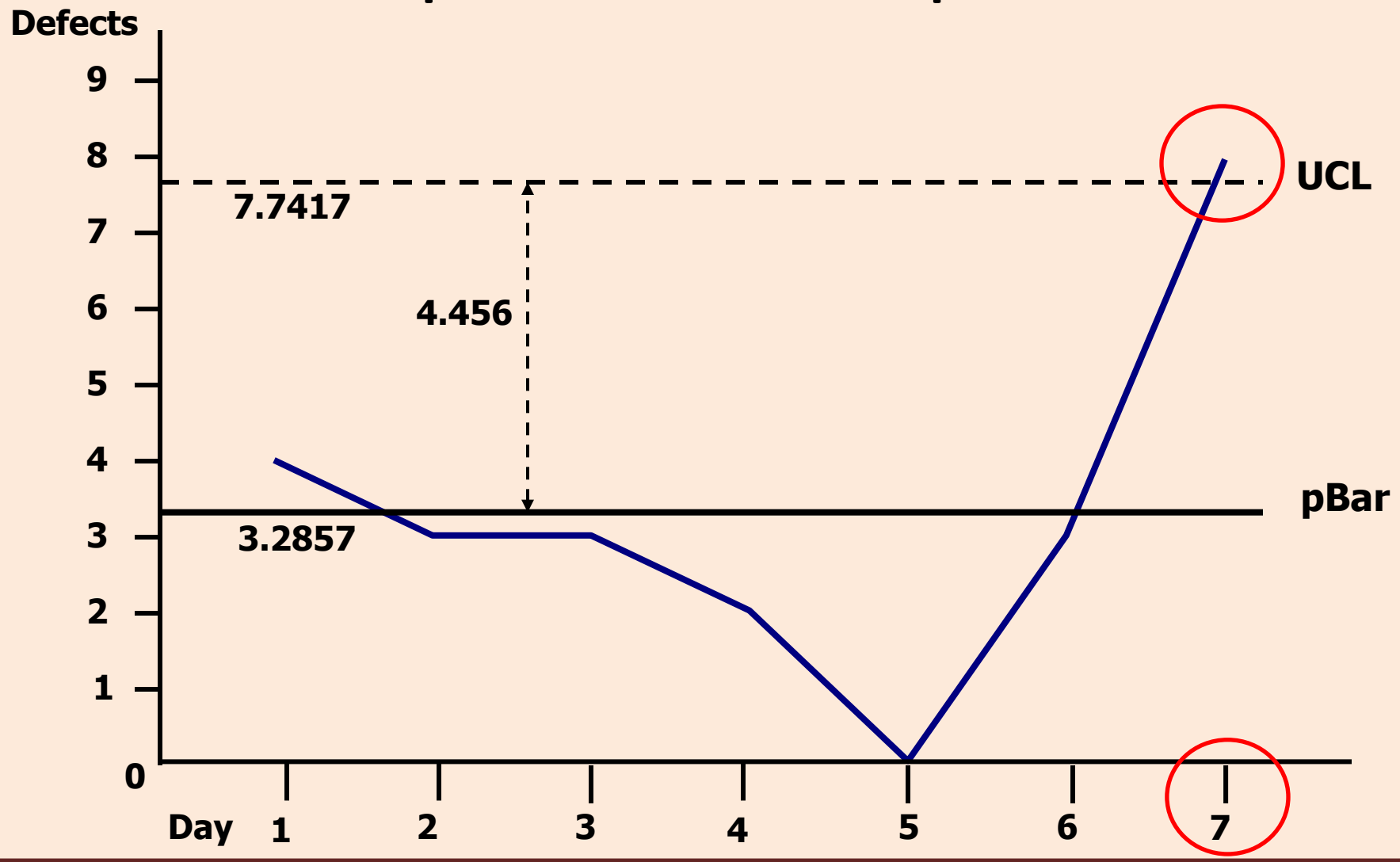
Example Using np Chart

- A bank produces daily account summaries for major clients
- Samples of 10 reports are taken for 7 days of operation
- Reports are inspected for defects (non-conformities)

Day	1	2	3	4	5	6	7
Defects	4	3	3	2	0	3	8

- Center Line = (4+3+3+2+0+3+8)/7 = 3.2857
- Proportion of Defects = 3.2857/10 = 0.32857
- Control Limits = $3.2857 \pm 3 \times \text{Sqrt}[3.2857 \times (1-0.32857)]$
- Upper Limit = 7.7417
- Lower Limit = 0 (because actual value is negative)

p-Chart Example



Six Sigma

- A six sigma process requires that the variation in the process be reduced to such an extent that there are six standard deviations between the target mean and the closest specification limit
- Probability of defect or error is outside 6 standard deviations on either side of the mean!!

Six Sigma as Applied to Misspellings

SIGMA	PPM (best case)	PPM (worst case)	Misspellings
1 sigma	317,400	697,700	170 words per page
2 sigma	45,600	308,733	25 words per page
3 sigma	2,700	66,803	1.5 words per page
4 sigma	64	6,200	1 word per 30 pages (1 per chapter)
5 sigma	0.6	233	1 word in a set of encyclopedias
6 sigma	0.002	3.4	1 in all of the books in a small library

Six Sigma Process

- DMAIC is an extension of TDQM
- D = Define the goals
- M = Measure current situation
- A = Analyze case of problems
- I = Make improvements
- C = Control the process (monitor)

Questions and Discussion